

Deriverse DEX Audit Summary

Prepared by Cyfrin

Version 2.0

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December 15, 2025

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Deriverse on Deriverse DEX. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated December 15, 2025, was delivered to Deriverse, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 25 days, the Cyfrin team conducted an audit on the Deriverse DEX code provided by Deriverse. The findings consist of 20 High, 37 Medium & 18 Low severity issues with the remainder being informational.

Summary

Project Name	Deriverse DEX
Repository	protocol-v1
Commit	1edfaa3c670e...
Fix Commit	871e06d12eaa...
Audit Timeline	Oct 29th - Dec 2nd, 2025
Methods	Manual Review, Formal Verification

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	20	20	0
Medium Risk	37	37	0
Low Risk	18	18	0
Informational	24	21	3
Gas Optimizations	0	0	0
Total	99	96	3

6 Audit Scope

The audit scope was limited to:

```
src/program/perp/leverage_validation.rs
src/program/perp/mod.rs
src/program/perp/perp_context.rs
src/program/perp/perp_engine.rs
src/program/processor/airdrop.rs
src/program/processor/change_ref_program.rs
src/program/processor/close_account.rs
src/program/processor/deposit.rs
src/program/processor/dividends_allocation.rs
src/program/processor/dividends_claim.rs
src/program/processor/fees_deposit.rs
src/program/processor/fees_withdraw.rs
src/program/processor/mod.rs
src/program/processor/move_spot_avail_funds.rs
src/program/processor/new_base_crncy.rs
src/program/processor/new_holder_account.rs
src/program/processor/new_instrument.rs
src/program/processor/new_operator.rs
src/program/processor/new_perp_order.rs
src/program/processor/new_ref_link.rs
```

```
src/program/processor/new_root_account.rs
src/program/processor/new_spot_order.rs
src/program/processor/new_token.rs
src/program/processor/next_voting.rs
src/program/processor/perp_change_leverage.rs
src/program/processor/perp_deposit.rs
src/program/processor/perp_forced_close.rs
src/program/processor/perp_mass_cancel.rs
src/program/processor/perp_order_cancel.rs
src/program/processor/perp_quotes_replace.rs
src/program/processor/perp_statistics_reset.rs
src/program/processor/perp_withdraw.rs
src/program/processor/reset_tree.rs
src/program/processor/set_instr_oracle_feed.rs
src/program/processor/set_instr_ready_for_perp_upgrade.rs
src/program/processor/spot_lp.rs
src/program/processor/spot_mass_cancel.rs
src/program/processor/spot_order_cancel.rs
src/program/processor/spot_quotes_replace.rs
src/program/processor/swap.rs
src/program/processor/upgrade_to_perp.rs
src/program/processor/voting.rs
src/program/processor/withdraw_spl_tokens.rs
src/program/processor/withdraw.rs
src/program/spot/engine.rs
src/program/spot/mod.rs
src/program/spot/orders.rs
src/program/spot/spot_context.rs
src/program/constants.rs
src/program/create_client_account.rs
src/program/helper.rs
src/program/instruction_data.rs
src/program/log.rs
src/program/mod.rs
src/program/settings.rs
src/state/perps/perp_infos.rs
src/state/perps/perp_trade_header.rs
src/state/spots/spot_account_header.rs
src/state/spots/spot_accounts.rs
src/state/spots/spot_infos.rs
src/state/candles.rs
src/state/client_community.rs
src/state/client_drv.rs
src/state/client_primary_account_header.rs
src/state/client_primary.rs
src/state/community_account_header.rs
src/state/community.rs
src/state/holder.rs
src/state/instrument.rs
src/state/mod.rs
src/state/pdf.rs
src/state/perps.rs
src/state/root.rs
src/state/spots.rs
src/state/token.rs
src/state/types.rs
src/utils/access_manager.rs
src/utils/from_account_info.rs
src/utils/mod.rs
src/utils/new_types.rs
src/lib.rs
src/state/perps/perp_accounts.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Deriverse is a decentralized exchange (DEX) built on Solana that combines automated market maker (AMM) liquidity pools with central limit order book (CLOB) functionality for both spot and perpetual futures markets.

Spot Trading uses a hybrid architecture where the constant-product AMM (k = asset_tokens × crncy_tokens) provides baseline liquidity while limit orders on an RB-tree based order book offer price improvement. Traders can execute market orders, place limit orders, or swap tokens directly against the AMM & CLOB. Liquidity providers earn a share of trading fees distributed proportionally from protocol-collected fees.

Perpetual Futures operates on an isolated margin model with configurable leverage (dynamically capped based on daily volatility). The system includes funding rate mechanisms that periodically rebalance long/short positions, automated margin calls with penalty rates feeding an insurance fund, and a rebalancing mechanism for position management. Traders must purchase a "market seat" at a dynamic price based on current participant count before trading perpetuals; this seat can be sold back when exiting. Perpetual markets are created by upgrading existing spot instruments when they meet readiness criteria.

Tokenomics centers on the DRVS native token, which enables participation in on-chain governance voting (fee rates, discount parameters, pool ratios) and earns dividend distributions from protocol fees. Additionally, users can receive airdrops according to their spot/perp trading and LP activities.

Additional Features include a referral program providing fee discounts to referred traders with rewards to referrers, a points program that accumulates trading/LP activity rewards convertible to DRVS via airdrops, and a private mode for controlled launches with whitelisted participants. All protocol state is stored on-chain through Solana PDAs.

9 Attestation

Cyfrin attests that it performed a security review of the Deriverse DEX code provided by Deriverse.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-12-15-cyfrin-deriverse-dex-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Deriverse has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=a588d0ebf3f9d425b4f74273c1e651db373cce7884a381cea3785477dd7a93b6
F=2025-12-15-cyfrin-deriverse-dex-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.